RC BEAM – BENDING DESIGN TO EN 1992-1-1

Edit the inputs in section 2 (or in the Parameters panel) – every result updates.

1. DESIGN BASIS

ULS, EN 1990 eq. 6.10. Singly reinforced rectangular section (EN 1992-1-1 §3.1.7).

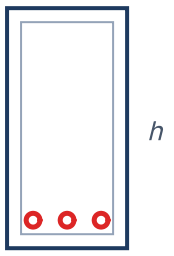
2. INPUTS

2.1 Materials

$f_{ck} := 30 \cdot \text{MPa}$ $\gamma_c := 1.5$ $\alpha_{cc} := 0.85$

$f_{yk} := 500 \cdot \text{MPa}$ $\gamma_s := 1.15$

2.2 Section



$b := 300 \cdot \text{mm}$ $h := 600 \cdot \text{mm}$ $c_{nom} := 35 \cdot \text{mm}$ $\phi_w := 10 \cdot \text{mm}$

$\phi := 20 \cdot \text{mm}$ $n := 6$ *tension bars of diameter ϕ*

2.3 Span and loads

$L := 7.2 \cdot \text{m}$ $g_k := 25 \cdot \frac{\text{kN}}{\text{m}}$ $q_k := 15 \cdot \frac{\text{kN}}{\text{m}}$

3. DESIGN ACTIONS

$$w_{Ed} := 1.35 g_k + 1.5 q_k = 56.25 \cdot \text{kN/m}$$

$$M_{Ed} := \frac{w_{Ed} L^2}{8} = \frac{(56.25 \cdot \text{kN/m}) \cdot (7.2 \cdot \text{m})^2}{8} = 364.5 \cdot \text{kN} \cdot \text{m}$$

$$V_{Ed} := \frac{w_{Ed} L}{2} = 202.5 \cdot \text{kN}$$

4. MATERIAL DESIGN VALUES

$$f_{cd} := \frac{\alpha_{cc} f_{ck}}{\gamma_c} = \frac{0.85 \cdot (30 \cdot \text{MPa})}{1.5} = 17 \cdot \text{MPa}$$

$$f_{yd} := \frac{f_{yk}}{\gamma_s} = 434.78 \cdot \text{MPa}$$

$$f_{ctm} := 0.30 \left(\frac{f_{ck}}{\text{MPa}} \right)^{\frac{2}{3}} \text{MPa} = 2.9 \cdot \text{MPa}$$

5. BENDING DESIGN

5.1 Effective depth

$$d := h - c_{nom} - \phi_w - \frac{\phi}{2} = 0.6 \cdot \text{m} - 0.035 \cdot \text{m} - 0.01 \cdot \text{m} - \frac{0.02 \cdot \text{m}}{2} = 545 \cdot \text{mm}$$

5.2 Lever arm

$$K_{Ed} := \frac{M_{Ed}}{b \cdot d^2 \cdot f_{ck}} = 0.136$$

$K_{lim} := 0.167$

$K_{\text{Ed}} = 0.136 \leq K_{\text{lim}} = 0.167$ PASS

No compression steel needed.

$$z := \min\left(d \left(0.5 + \sqrt{0.25 - \frac{K_{Ed}}{1.134}} \right), 0.95 d \right) = 469 \cdot \text{mm}$$

5.3 Tension reinforcement

$$A_{s.req} := \frac{M_{Ed}}{f_{yd} z} = 1788 \cdot \text{mm}^2$$

$$A_{s.prov} := n \frac{\pi \phi^2}{4} = 1885 \cdot \text{mm}^2$$

$$\eta_M := \frac{A_{s.req}}{A_{s.prov}} = 0.949$$

$\eta_M = 0.949 \leq 1$ PASS

6. MINIMUM REINFORCEMENT (§9.2.1.1)

$$A_{s.min} := \max\left(0.26\,\frac{f_{ctm}}{f_{yk}}\,b \cdot d,\, 0.0013 b \cdot d\right) = 246 \cdot \text{mm}^2$$

$$A_{\text{s.min}} = 246.26 \cdot \text{mm}^2 \;\leq\; A_{\text{s.prov}} = 1884.96 \cdot \text{mm}^2$$ PASS

7. SUMMARY

$M_{Ed} = 364.5 \cdot \text{kN} \cdot \text{m}$ $A_{s.prov} = 1885 \cdot \text{mm}^2$ $\eta_M = 0.949$